Comparing Methods for Integrating Personalized eXtended Reality in Stroke Rehabilitation

Inês Figueiredo · IEETA, DETI, University of Aveiro, Portugal · *ines.figueiredo@ua.pt*
Bernardo Marques · IEETA, DETI, University of Aveiro, Portugal · *bernardo.marques@ua.pt*
David Palricas · IEETA, DETI, University of Aveiro, Portugal · *davidpalricas@ua.pt*
Sérgio Oliveira · IEETA, DETI, University of Aveiro, Portugal · *sergiodoliveira@ua.pt*
Bianca Guerreiro · IEETA, DETI, University of Aveiro, Portugal · *bianca.guerreiro@ua.pt*
Pedro Reisinho · IEETA, DETI, University of Aveiro, Portugal · *pedro.reisinho@ua.pt*
Samuel Silva · IEETA, DETI, University of Aveiro, Portugal · *sss@ua.pt*
Paulo Dias · IEETA, DETI, University of Aveiro, Portugal · *paulo.dias@ua.pt*
Paula Amorim · CMRRC-Rovisco Pais, Portugal · *ploamorim@gmail.com*
Beatriz Sousa Santos · IEETA, DETI, University of Aveiro, Portugal · *bss@ua.pt*

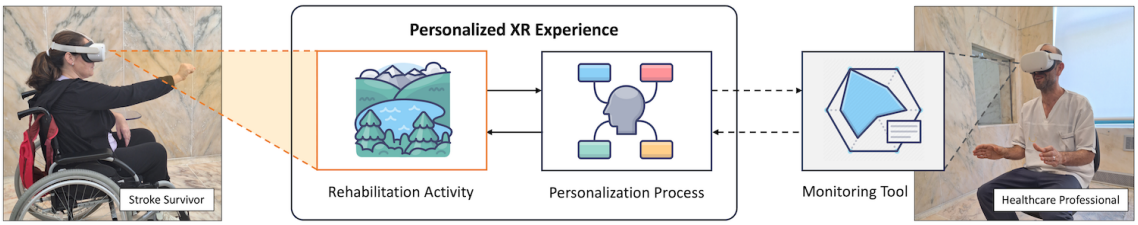


Figure 1. Rehabilitation scenario with a stroke survivor (left) supported by a Personalized XR experience and a healthcare professional (right), with access to a monitoring tool, capable of adjusting the rehabilitation process as needed. Assets from https://iconfinder.com.

Abstract. Stroke results in long-term motor and cognitive impairments, making rehabilitation essential for the recovery of quality of life. However, traditional approaches lack personalization, limiting their ability to adapt to the need/progress of each individual. Recent advances in eXtended Reality (XR) offer promising avenues to deliver tailored, engaging experiences. These personalized environments can be configured automatically through adaptive algorithms or manually by healthcare professionals, enabling more responsive and motivating rehabilitation sessions. This means that even when rehabilitation is delivered individually, the process is inherently social, requiring ongoing involvement from professionals, which means that Personalized XR must be understood as extending beyond a single isolated activity. This work identifies, compares, and discusses existing methods for integrating personalization into XR-based stroke rehabilitation. By analyzing distinct characteristics, the study aims to highlight current strengths, limitations, and opportunities for improvement. Ultimately, enhancing personalization in XR rehabilitation could increase survivors' adherence, motivation, and overall quality of life.

Keywords. Personalization, Monitoring, eXtended Reality, Virtual Reality, Augmented Reality, Serious Game, Stroke, Rehabilitation

---

Inês Figueiredo, Bernardo Marques, David Palricas, Sérgio Oliveira, Bianca Guerreiro, Pedro Reisinho, Samuel Silva, Paulo Dias, Paula Amorim, Beatriz Sousa Santos. 2026. Comparing Methods for Integrating Personalized eXtended Reality in Stroke Rehabilitation. In: *Proceedings of the 24th EUSSET Conference on Computer-Supported Cooperative Work (ECSCW) – Posters and Demos*, Reports of the European Society for Socially Embedded Technologies. ISSN: 2510-2591 https://doi.org/10.48340/to-be-added

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers, or to redistribute to lists, contact the Authors.

Find the latest version of this document in the EUSSET Digital Library: https://dl.eusset.eu/

1 Introduction

Stroke leads to physical and cognitive impairments, significantly altering survivors' ability to perform daily activities and maintain independence (Feigin et al., 2014; Negrini et al., 2020). These changes can affect movement, coordination, memory, attention, and communication, ultimately reducing quality of life and increasing the need for continuous care and support (Figueiredo et al., 2025a; Liang et al., 2021; Marques et al., 2026). To help survivors regain functionality and improve their overall well-being, rehabilitation plays a crucial role (Andreikanich et al., 2019; Oliveira et al., 2025c; Paraense et al., 2022). Through targeted and repetitive exercises, rehabilitation aims to restore motor control, cognitive performance, and confidence, supporting individuals in re-engaging with everyday life activities and social participation (Marques et al., 2024a; Oliveira et al., 2025b).

Despite their importance, traditional rehabilitation programs rely on standardized, one-size-fits-all approaches (Figueiredo et al., 2025c; Patsaki et al., 2022). Such methods fail to consider the unique abilities, progress rates, and motivational factors of each stroke survivor, limiting engagement and the effectiveness of therapy outcomes (Figueiredo et al., 2025b; Oliveira et al., 2024). Therefore, exploring more personalized rehabilitation methods is essential to address the diverse needs of stroke survivors. Integrating adaptive and human-centered strategies can make therapy more meaningful, motivating, and aligned with individual goals and recovery trajectories (Cucinella et al., 2025; Figueiredo et al., 2025d; Oliveira et al., 2025a). Moreover, rehabilitation should be understood as extending beyond a single isolated activity, as the process is inherently social and requires ongoing involvement from professionals.

To address this opportunity, this work focuses on personalized eXtended Reality (XR) rehabilitation experiences designed to assist stroke survivors. Specifically, it identifies, compares, and discusses existing approaches for incorporating personalization features into XR environments. By analyzing their characteristics, strengths, and limitations, this study highlights opportunities for advancement. Ultimately, the goal is to enhance how personalization in XR rehabilitation supports survivor motivation, engagement, and adherence—contributing to improved recovery and quality of life beyond the rehabilitation center.

2 Methods and Materials

This section outlines the rationale underlying this research line, emphasizing the growing importance of personalization within XR for stroke rehabilitation.

Using XR has emerged as a promising approach for stroke rehabilitation, offering controlled, immersive, and interactive environments that can assist physical and cognitive recovery. Literature highlights how XR can enhance engagement, support social and cognitive training, thereby complementing traditional therapeutic approaches (Barros et al., 2025; Marques et al., 2023, 2025). Beyond these, XR also holds potential for personalization, allowing experiences to be dynamically tailored to each individual's abilities, needs, and progress throughout the rehabilitation process. Such personalization can enable improved motivation, engagement, and adherence, ultimately contributing to more effective long-term recovery trajectories

(Liang et al., 2021; Marques et al., 2024b; Patsaki et al., 2022).

To achieve this vision, our team began by conducting a scoping review of the last five years (Figueiredo et al., 2025c) to understand current personalization methods in XR-based rehabilitation, identify emerging trends, and uncover gaps that limit clinical adoption. In parallel, we carried out a series of structured brainstorming sessions and on-site visits to a renowned rehabilitation center, engaging directly with healthcare professionals and stroke survivors. Guided by a Human-Centered Design (HCD) methodology (Figueiredo et al., 2025a), these activities enabled us to capture the needs, expectations, and challenges of the target population, providing concrete insights into how XR systems must adapt during rehabilitation to remain clinically relevant, supportive, and personalized.

To showcase a typical scenario, Figure 1 presents a conceptual model depicting a stroke survivor (left) engaged in a Personalized XR experience, while a healthcare professional (right) oversees progress through a dedicated monitoring interface. This illustrates how personalized adjustments, whether automated or manually-guided can be integrated into the rehabilitation workflow, enabling the XR experience to evolve alongside the survivor's recovery and ensuring that therapeutic challenges remain appropriate, supportive, and clinically aligned.

Regarding the Personalized XR experience, it can be further detailed through its underlying pipeline (see Figure 2). This pipeline begins with the XR Environment, where stroke survivors engage with interactive virtual rehabilitation scenarios designed to promote motor and cognitive training. Their interactions are continuously captured and routed through the Data Hub, where the information is processed, analyzed, and compared against previous sessions to track progress and detect changes over time. The Personalization Module then leverages these insights to adapt elements of the rehabilitation experience, adjusting difficulty, feedback, or task structure to better match the survivor's evolving abilities. Finally, the Monitoring Tool provides healthcare professionals with an accessible overview of the collected data, enabling them to interpret progress and make informed decisions about tailoring the intervention further, ultimately ensuring that expert judgment remains central to the rehabilitation process.

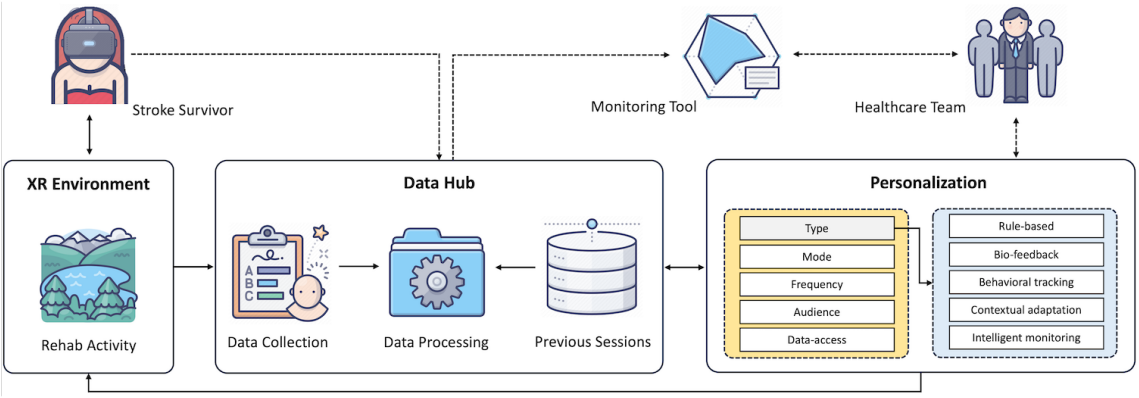


Figure 2. Representation showcasing a personalized XR-based pipeline for stroke rehabilitation. Assets from https://iconfinder.com.

Focusing in specific on the personalization module, and according to our scoping review (Figueiredo et al., 2025b,c), the field currently adopts five primary personalization types, namely:

- **Rule-based Systems** - Predefined rules adjust the XR experience based on performance data. For instance, if specific task milestones are unmet, the system can lower task difficulty or enhance assistance moving forward;
- **Bio-feedback** - Physiological data from sensors (e.g., heart rate, muscle activity) can drive personalization. High stress or fatigue levels might trigger a calming environment or a task intensity reduction;
- **Behavioral Tracking** - Monitoring movement, posture, and engagement informs when users face frustration or fatigue, enabling task simplifications, motivational boosts, or gamified enhancements;
- **Contextual Adaptation** - External factors like session timing, previous session duration, or user mood enable dynamic adjustments to session structure or intensity;
- **Intelligent Monitoring** - Intelligent algorithms and techniques can analyze data patterns to predict user needs, allowing the system to refine task difficulty and improve its responsiveness over time.

Figure 3 illustrates a conceptual pipeline for the XR personalization process, starting on the left with the inputs that guide personalization: data collected during the current session, information from previous sessions, and insights provided by healthcare professionals. After the appropriate 'Personalization Type' is considered, the process advances to 'Session Understanding', where the system organizes and summarizes stroke survivors' interactions. This is followed by 'Data Interpretation', in which the collected information is analyzed to extract meaningful patterns. The interpreted data is then processed by a 'Decision Engine' that determines what adaptations are appropriate. Finally, the 'Recommendation Generator' produces adaptation commands that are sent to the XR Environment on the right, resulting in adjustments to the rehabilitation activity experienced by the stroke survivors.

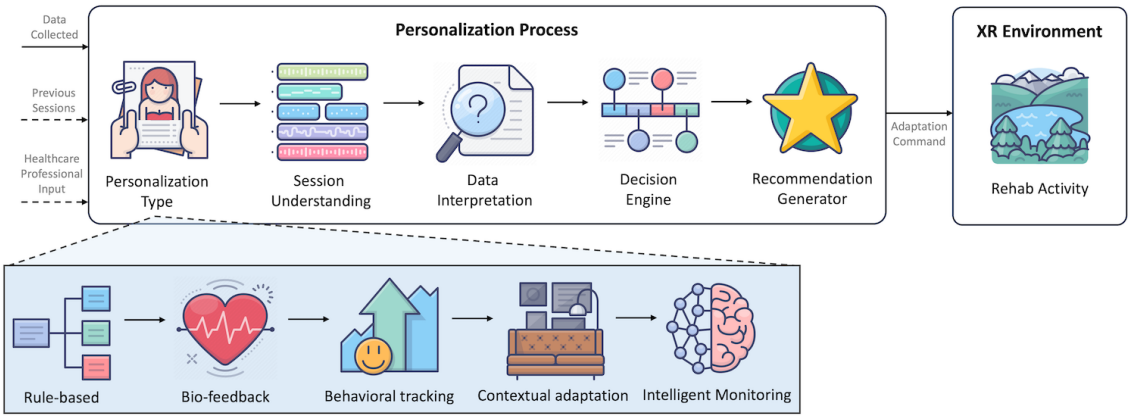


Figure 3. Overview of the XR personalization process, beginning with data inputs and the selection of a personalization type. The process then advances through session understanding, data interpretation, and decision making to generate adaptation commands. These commands update the XR environment, tailoring the rehabilitation activity experienced by stroke survivors. Assets from https://iconfinder.com.

3 Comparison and Discussion

This section examines existing methods for implementing Personalized XR in rehabilitation with a focus on stroke recovery. Table 1 presents a comparison of the the main advantages/disadvantages, based on the authors experience in the field, as well as our previous scoping review, were we payed close attention to the personalization type being considered, as well as any details regarding its implementation and possible limitations. With this, we intend to identify both the strengths that enable effective personalization and the limitations that hinder broader clinical adoption of these technological approaches to rehabilitation.

Table 1. Comparison of different methods for Personalized XR-based stroke rehabilitation.

Method	Advantages	Disadvantages
Rule-based	• Simple to implement and explain • Predictable behavior - easy for clinicians to control • Low computational requirements	• Limited adaptability to complex or unexpected behavior • Requires manual rule creation • May oversimplify user needs and fail in nuanced or complex rehabilitation scenarios
Bio-feedback	• Captures user involvement grounded in physiological state • Can address stress, fatigue, and mental load directly • Objective data reduces reliance on user self-report	• Requires reliable sensors and calibration (more intrusive) • Susceptible to noise, artifacts, and measurement errors • Higher cost and setup burden for clinical environments
Behavioral tracking	• Captures detailed movement and engagement patterns • Provides rich data for monitoring and analyzes	• Data processing can be computationally intensive • Accuracy depends on tracking hardware and environment • Risk of misinterpretation (e.g., attributing poor performance to fatigue instead of motor deficits)
Contextual adaptation	• Adapts experiences based on broader situational factors • Aligns therapy intensity with daily fluctuations in energy or emotional state	• Data processing can be computationally intensive • Difficult to quantify certain contextual variables • Can rely on assumptions rather than direct evidence
Intelligent monitoring	• Enables predictive, proactive personalization • Learns complex patterns across sessions • Potential for highly adaptive, individualized rehabilitation plans	• Requires large, high-quality datasets to train models • Data processing can be highly computationally intensive • Lower transparency/interpretability - clinicians may distrust “black-box” behavior • Harder to validate clinically and integrate into regulated environments

Next, we elaborate on the data from Table 1 by highlighting how each method offers distinct strengths and limitations. The insights derived are intended to guide future developments, encouraging the community to pursue more transparent, interoperable, and clinically grounded strategies for integrating personalized XR.

- **Rule-based** personalization considered is easy to design and implement, requiring no complex algorithms or large datasets. Its predictable behavior allows healthcare professionals to understand and control how the XR system adapts, helping them integrate it confidently into therapy. Because it demands minimal computational resources, it runs reliably on standard hardware, making it practical for most rehabilitation settings (Afyouni et al., 2019; Darzi et al., 2021; Han et al., 2024). However, rule-based systems adapt poorly to unpredictable behavior and cannot capture subtle changes common in stroke recovery. Because these systems rely on predefined logic, they cannot easily adjust to variations in user performance or progress. Plus, creating and maintaining effective rules requires manual effort by experts, and the logic can quickly become restrictive as rehabilitation needs grow more complex (Lee and Stuerzlinger, 2025).

- **Bio-feedback** captures user involvement directly from physiological data, enabling adaptations that reflect the survivor's actual stress, fatigue, or cognitive load. This helps the system react to conditions that may not be visible through behavior alone. Because the data is objective, it reduces reliance on self-report, which can be inconsistent or difficult for some stroke survivors to communicate accurately (Lopes and Lopes, 2022; Zhao et al., 2025). However, bio-feedback requires reliable sensors and regular calibration, making it more intrusive and less convenient in clinical settings. Physiological signals are often noisy and prone to artifacts, which can lead to inaccurate or unstable adaptations. The need for additional equipment also increases cost and setup time, creating a heavier operational burden for rehabilitation centers (Guillen-Sanz et al., 2024; Lüddecke and Felnhofer, 2022).

- **Behavioral tracking** captures detailed movement and engagement patterns, offering a rich understanding of motor performance that is highly relevant for stroke rehabilitation. It provides comprehensive data that healthcare professionals can use to monitor progress and analyze how survivors perform tasks within the virtual environment, enabling more informed adjustments to therapy plans (Huang et al., 2025). Despite its strengths, behavioral tracking can be computationally demanding, especially when processing full-body or high-frequency motion data. Its accuracy depends heavily on the quality of tracking hardware and environmental conditions, which can introduce errors or inconsistencies. Additionally, behavioral data may be misinterpreted. For example, mistaking motor impairment for fatigue, leading to inappropriate adaptations if not carefully processed (Michiels et al., 2024).

- **Contextual adaptation** adapts experiences based on meaningful situational factors, allowing the XR system to better match the survivor's readiness and daily fluctuations in energy or emotional state. By accounting for these broader influences, it can help optimize engagement and prevent overload, supporting a smoother and more personalized rehabilitation process (Liu et al., 2011; Lo et al., 2022). However, contextual adaptation can be computationally demanding when integrating diverse sources of contextual data. Many contextual variables, such as mood or mental fatigue are difficult to quantify reliably, which may force the

system to rely on assumptions rather than direct evidence. As a result, adaptations may be less precise than those driven by performance or physiological sensing (Valmorisco et al., 2024).

- **Intelligent monitoring:** This approach enables predictive and proactive personalization, allowing the system to intervene before performance issues arise. It can learn complex patterns across sessions, capturing subtle changes that simpler methods may overlook. As a result, intelligent monitoring offers the potential for highly adaptive, individualized rehabilitation plans that evolve continually as the survivor recovers (Avola et al., 2019; Lee et al., 2021; Mazzolenis et al., 2024). Despite its promise, intelligent monitoring requires large, high-quality datasets to train reliable models, a challenge for many rehabilitation centers with limited resources. The data processing demands are high, making real-time integration computationally intensive. Its 'black-box' nature also reduces transparency and interpretability, which may lead healthcare professionals to distrust automated decisions. Moreover, this approach is harder to validate clinically and integrate into regulated healthcare environments, slowing their adoption (Abedi et al., 2024; Linse et al., 2022; Ribeiro de Oliveira et al., 2023).

4 Concluding Remarks and Future Work

Literature shows that XR-solutions for stroke rehabilitation remains centered on standardized, isolated experiences that do not fully account for individual variability in recovery needs and capabilities. This can reduce engagement, hinder optimal recovery, and overlook critical aspects of personalized care. Even when rehabilitation is delivered individually, the process is inherently social, requiring involvement from healthcare professionals, which means that Personalized XR must be understood as extending beyond a single isolated activity. Moreover, healthcare professionals lack adequate tools to monitor rehabilitation progress and tailor XR experience parameters to each individual's evolving needs, preferences, and limitations.

This work identifies, compares, and discusses existing methods for integrating personalization into XR–based stroke rehabilitation. By examining a range of approaches, it aims to highlight the current strengths, limitations, and opportunities that shape the development of Personalized XR systems. Our analysis shows that rule-based systems stand out as one of the most frequently adopted approaches due to their simplicity, lower computational requirements, and ease of implementation, which likely explains their strong presence despite their limited ability to react to unexpected behavior. More recently, intelligent monitoring has begun to surface, yet its practical adoption remains minimal because it depends on large, high-quality datasets and advanced hardware, resources that many rehabilitation centers lack. Lastly, although bio-feedback offers valuable real-time insights into the survivor's physiological state, its need for additional sensors, calibration, and susceptibility to noise makes it less convenient in practice, not due to weaknesses in the personalization itself but because of the added setup burden and potential discomfort or distraction for survivors.

Moving forward, future research should aim to to design and disseminate an online questionnaire targeting researchers and target users involved in XR-rehabilitation. Gathering insights from diverse stakeholders could reveal practical challenges, preferences, and success factors that are not yet well-documented in the literature. Another promising direction involves the development of standardized evaluation frameworks for assessing the effectiveness and robustness of personalization methods in XR rehabilitation. Future work should investigate which metrics are most appropriate for specific rehabilitation scenarios. Establishing clear validation procedures would ensure that personalized XR can be objectively compared, thereby improving their reliability and rehabilitation relevance.

Acknowledgments

We thank everyone involved from the Rovisco Pais rehabilitation center for their time and expertise. This work was funded by European Regional Development Fund (FEDER) under COMPETE 2030/Portugal 2030 Program (COMPETE2030-FEDER-00815600/#16874), and was supported by the Portuguese Foundation for Science and Technology (FCT) under the IC&DT 2023 Call. This work was supported by FCT through contract doi.org/10.54499/UID/00127/2025 - IEETA - Institute of Electronics and Informatics Engineering of Aveiro.

References

Abedi, Ali, Tracey J. F. Colella, Maureen Pakosh, and Shehroz S. Khan. 2024. Artificial intelligence-driven virtual rehabilitation for people living in the community: A scoping review. *npj Digital Medicine*, 7 (1): 25.

Afyouni, Imad, Anas Einea, and Abdullah Murad. 2019. RehaBot: Gamified Virtual Assistants Towards Adaptive TeleRehabilitation. In: *Adjunct Publication of the 27th Conference on User Modeling, Adaptation and Personalization*, p. 21–26.

Andreikanich, Anna, Beatriz Sousa Santos, Paula Amorim, Helder Zagalo, Bernardo Marques, Paulo Margalho, Jorge Laíns, Fátima Faim, Margarida Coelho, Teresa Cardoso, et al. 2019. An Exploratory Study on the use of Virtual Reality in Balance Rehabilitation. In: *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 3416–3419.

Avola, Danilo, Luigi Cinque, Gian Luca Foresti, and Marco Raoul Marini. 2019. An interactive and low-cost full body rehabilitation framework based on 3D immersive serious games. *Journal of Biomedical Informatics*, 89: 81–100.

Barros, Fábio, Bernardo Marques, Samuel Silva, António Teixeira, and Beatriz Sousa Santos. 2025. A vision of extended reality as an off-the-shelf modality in multimodal interactive ecosystems. In: *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1059–1064.

Cucinella, Salvatore Luca, Joost CF de Winter, Erik Grauwmeijer, Marc Evers, and Laura Marchal-Crespo. 2025. Towards personalized immersive virtual reality neurorehabilitation: a human-centered design. *Journal of NeuroEngineering and Rehabilitation*, 22 (1): 7.

Darzi, Ali, Sean M McCrea, and Domen Novak. 2021. User Experience With Dynamic Difficulty Adjustment Methods for an Affective Exergame: Comparative Laboratory-Based Study. *JMIR Serious Games*, 9 (2).

Feigin, Valery, Mohammad Forouzanfar, Rita Krishnamurthi, George Mensah, Myles Connor, Derrick Bennett, Andrew Moran, Ralph Sacco, Laurie Anderson, and Thomas Truelsen. 2014. Global and regional burden of stroke during 1990-2010: findings from the Global Burden of Disease Study. *The Lancet*, 383 (9913): 245–255.

Figueiredo, Inês, Bernardo Marques, Sérgio Oliveira, Paula Amorim, and Beatriz Sousa Santos. 2025a. What do Healthcare Professionals Need? An Online Study on Personalized Virtual Reality for Stroke

Recovery at a Rehabilitation Center. In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 1–8.

Figueiredo, Inês, Bernardo Marques, Sérgio Oliveira, Paulo Dias, and Beatriz Sousa Santos. 2025b. Towards Adaptive Virtual Reality for Stroke Recovery: A Personalized Approach to Rehabilitation. In: *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1046–1050.

Figueiredo, Inês, Bernardo Marques, Sérgio Oliveira, Bianca Guerreiro, Samuel Silva, Paula Amorim, Tiago Araújo, Liliana Vale Costa, Carlos Ferreira, Paulo Dias, et al. 2025c. Personalized eXtended Reality experiences to enhance the rehabilitation process of stroke survivors: A scoping review. *Computers & Graphics*, pp. 1–16.

Figueiredo, Inês, Sérgio Oliveira, Bernardo Marques, Paula Amorim, and Beatriz Sousa Santos. 2025d. ADAPT - Empowering Healthcare Professionals to Tailor Collaborative VR-Games for Stroke Recovery. In: *IEEE Conference on Games (CoG)*, pp. 1–4.

Guillen-Sanz, Henar, David Checa, Ines Miguel-Alonso, and Andres Bustillo. 2024. A systematic review of wearable biosensor usage in immersive virtual reality experiences. *Virtual Reality*, 28 (2): 74.

Han, Renzhi, Boon Giin Lee, Dave Towey, Yuan Yao, and Matthew Pike. 2024. Exploring Emotional Responses with Dynamic Difficulty Adjustment Adaptation in Immersive Virtual Reality Exergaming. In: *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 621–627.

Huang, Nan, Prashant Goswami, Veronica Sundstedt, Yan Hu, and Abbas Cheddad. 2025. Personalized smart immersive XR environments: a systematic literature review. *Vis. Comput.*, 41 (11): 8593–8626.

Lee, Jong-In and Wolfgang Stuerzlinger. 2025. Towards Personalized Navigation in XR: Design Recommendations to Accommodate Individual Differences. In: *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 241–245.

Lee, Si-Huei, Jianjun Cui, Lizheng Liu, Mu-Chun Su, Lirong Zheng, and Shih-Ching Yeh. 2021. An Evidence-Based Intelligent Method for Upper-Limb Motor Assessment via a VR Training System on Stroke Rehabilitation. *IEEE Access*, 9: 65871–65881.

Liang, Ci-Jyun, Charles Start, Hanna Boley, Vineet R Kamat, Carol C Menassa, and Michelle Aebersold. 2021. Enhancing stroke assessment simulation experience in clinical training using augmented reality. *Virtual Reality*, 25: 575–584.

Linse, Christoph, Hammam Alshazly, and Thomas Martinetz. 2022. A walk in the black-box: 3D visualization of large neural networks in virtual reality. *Neural Computing and Applications*, 34 (23): 21237–21252.

Liu, Wei, Xue Li, and Daoli Huang. 2011. A survey on context awareness. In: *2011 International Conference on Computer Science and Service System (CSSS)*, pp. 144–147.

Lo, Wei Hong, Holger Regenbrecht, Barrett Ens, and Stefanie Zollmann. 2022. A Context-aware Interface for Immersive Sports Spectating. In: *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 605–610.

Lopes, Júlio Castro and Rui Pedro Lopes. 2022. A Review of Dynamic Difficulty Adjustment Methods for Serious Games. In: Ana I. Pereira, Andrej Košir, Florbela P. Fernandes, Maria F. Pacheco, João P. Teixeira, and Rui P. Lopes (eds.): *Optimization, Learning Algorithms and Applications*. Cham: Springer International Publishing.

Lüddecke, Robin and Anna Felnhofer. 2022. Virtual Reality Biofeedback in Health: A Scoping Review. *Applied Psychophysiology and Biofeedback*, 47 (1): 1–15.

Marques, Bernardo, Carlos Ferreira, Samuel Silva, Paulo Dias, and Beatriz Sousa Santos. 2023. Is social presence (alone) a general predictor for good remote collaboration? comparing video and augmented reality guidance in maintenance procedures. *Virtual Reality*, 27 (3): 1783–1796.

Marques, Bernardo, Carlos Ferreira, Samuel Silva, André Santos, Andreia Santos, Paulo Dias, and Beatriz Sousa Santos. 2025. Exploring different content creation and display methods for remote collaboration supported by eXtended reality: Comparative analysis of distinct task scenarios. *Multimedia Tools and Applications*, 84 (18): 19785–19815.

Marques, Bernardo, Sérgio Oliveira, Hugo Ferreira, Paula Amorim, Paulo Dias, and Betriz Sousa Santos. 2024a. The Role of Collaborative Virtual Reality Engagement in Stroke Survivors' Rehabilitation. *Ac-*

cessibility, Assistive Technology and Digital Environments. International Conference on Applied Human Factors and Ergonomics (AHFE), 121.

Marques, Bernardo, Sérgio Oliveira, Inês Figueiredo, Paula Amorim, Mariana Leite, Carlos Ferreira, Samuel Silva, David Palricas, Paulo Dias, and Beatriz Sousa Santos. 2026. Leveraging Social Interaction: Stroke Rehabilitation Using eXtended Reality. *IEEE Computer Graphics and Applications*, 46 (2): 131–140.

Marques, Bernardo, Samuel Silva, and Betriz Sousa Santos. 2024b. Measuring Remote Collaboration Supported by Extended Reality: A Multidimensional Evaluation Approach. *Accessibility, Assistive Technology and Digital Environments. International Conference on Applied Human Factors and Ergonomics (AHFE)*, 121.

Mazzolenis, Maria Victoria, Gabrielle Naime Mourra, Sacha Moreau, Maria Emilia Mazzolenis, Ivo H. Cerda, Julio Vega, James S. Khan, and Alexandra Thérond. 2024. The Role of Virtual Reality and Artificial Intelligence in Cognitive Pain Therapy: A Narrative Review. *Current Pain and Headache Reports*, 28 (9): 881–892.

Michiels, Nick, Lode Jorissen, Jeroen Put, Jori Liesenborgs, Isjtar Vandebroeck, Eric Joris, and Frank Van Reeth. 2024. Tracking and co-location of global point clouds for large-indoor environments. *Virtual Reality*, 28 (2): 106.

Negrini, Stefano, Klemen Grabljevec, Paolo Boldrini, Carlotte Kiekens, Sasa Moslavac, Mauro Zampolini, Nicolas Christodoulou, and Jean-François Kaux. 2020. Up to 2.2 million people experiencing disability suffer collateral damage each day of COVID-19 lockdown in Europe. *European Journal of Physical and Rehabilitation medicine*, 56 (3): 361–365.

Oliveira, Sérgio, Inês Figueiredo, Bernardo Marques, Paula Amorim, and Beatriz Sousa Santos. 2025a. REVIVE - Studying Social Dynamics of Stroke Survivors using a Monitoring VR-based Platform. In: *IEEE Conference on Games (CoG)*, pp. 1–4.

Oliveira, Sérgio, Bernardo Marques, Paula Amorim, Paulo Dias, and Betriz Sousa Santos. 2024. Beyond the Prototype: Extended Use of a Virtual Reality Supermarket in a Rehabilitation Center. In: *International Conference on Graphics and Interaction (ICGI).*, pp. 1–8.

Oliveira, Sérgio, Bernardo Marques, Paula Amorim, Inês Figueiredo, Bianca Guerreiro, David Palricas, Samuel Silva, Paulo Dias, and Beatriz Sousa Santos. 2025b. Exploring Social Dynamics through Multi-User Virtual and Augmented Reality Serious Games for Stroke Rehabilitation. In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 1–4.

Oliveira, Sérgio, Bernardo Marques, Paula Amorim, and Beatriz Sousa Santos. 2025c. Collaborative Virtual Reality Serious Games as a Therapeutic Medium for Stroke Survivors at a Rehabilitation Center. In: *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1051–1058.

Paraense, Hélder, Bernardo Marques, Paula Amorim, Paulo Dias, and Beatriz Sousa Santos. 2022. Whac-A-Mole: Exploring virtual reality (VR) for upper-limb post-stroke physical rehabilitation based on participatory design and serious games. In: *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 716–717.

Patsaki, Irini, Nefeli Dimitriadi, Akylina Despoti, Dimitra Tzoumi, Nikolaos Leventakis, Georgia Roussou, Argyro Papathanasiou, Serafeim Nanas, and Eleftherios Karatzanos. 2022. The effectiveness of immersive virtual reality in physical recovery of stroke patients: A systematic review. *Frontiers in Systems Neuroscience*, 16.

Ribeiro de Oliveira, Tainã, Brenda Biancardi Rodrigues, Matheus Moura da Silva, Rafael Antonio N. Spinassé, Gabriel Giesen Ludke, Mateus Ruy Soares Gaudio, Guilherme Iglesias Rocha Gomes, Luan Guio Cotini, Daniel da Silva Vargens, Marcelo Queiroz Schimidt, Rodrigo Varejão Andreão, and Mário Mestria. 2023. Virtual Reality Solutions Employing Artificial Intelligence Methods: A Systematic Literature Review. *ACM Comput. Surv.*, 55 (10).

Valmorisco, Sergio, Laura Raya, and Alberto Sanchez. 2024. Enabling personalized VR experiences: a framework for real-time adaptation and recommendations in VR environments. *Virtual Reality*, 28 (3): 128.

Zhao, Yuchen, Jay David Bolter, Kexin Chen, Xuerui Wang, Xintong Jiang, and Zhao Ni. 2025. BioVR: An Exploratory Study of Biofeedback-Driven Adaptive VR for Personalized and Sustained Fitness Intensity. In: *Companion Publication of the 2025 ACM Designing Interactive Systems Conference*, p. 586–591.